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NataliaYakushkina
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Natalia Yakushkina

B.Sc. student in Computer Visualization & Design, focused on **real-time 3D**, **XR** and **digital twins**. Six months at **Fraunhofer IEM** on a live-data airport digital twin in VR; bachelor's thesis on **LLM-driven AI guides** in 3D environments. Background in 3D modeling and UX design, plus ten years of founding and running a company.

Experience

Apr 2025 – Sep 2025 **Digital Twin & XR Intern**, *Fraunhofer IEM (AR/VR Lab)*, Paderborn, Germany, Demo

Project FastGate: interactive Unity digital twin of apron processes at Paderborn/Lippstadt Airport.

- Integrated live sensor data via Apache Kafka.
- Adapted the app for VR: locomotion, VR UI, controller hints.
- Built live-data tooltips for moving objects.
- Modeled and rigged an animated 3D airport worker.

Sep 2010 – Jul 2020 **Co-Founder, Baltik Bread**, *Bakery Manufacturing & Mobile Retail*, Novosibirsk, Russia

- Co-founded a manufacturing + retail business; 500–600k EUR investment, full ROI in 4 years.
- Grew to ∼1M EUR annual revenue, ∼2,500 units/day across ∼40 SKUs with 90%+ sell-through.
- Planned daily production, raw-material flow and delivery logistics for 6 sales points and wholesalers.
- Hired, trained and led a 25-person team; created onboarding and work procedures.

Projects

2026 **Bachelor's Thesis: Spatial Awareness for AI Avatars**, *Unity, Python*, Demo · Code

- Designed an LLM guide for a Unity art gallery using a scene graph and GPT-4o-mini tool calling.
- Reached 96% accuracy on 48 spatial queries.

2026 **Household Helper Web App**, *React, Node.js, SQLite*, Demo · Code

- Built the React PWA front end in a 3-person Scrum team using GitHub.

2025 **AR App for Urban Greening in Lippstadt**, *Unity, Vuforia, Blender*, Demo · Code

- Created a mobile AR app to place and adjust virtual trees outdoors via ground-plane detection.
- Optimized 3D tree assets for mobile and added wind animation.

2024 **Virtual Try-On Glasses App**, *Unity, AR Foundation*, Demo

- Developed an AR glasses try-on with face tracking and skin tone customization.

Education

Oct 2020 – Present **B.Sc. Computer Visualization & Design**, *Hochschule Hamm-Lippstadt*, Lippstadt, Germany

Courses: Augmented Reality, 3D Modeling, Software Development, UX/UI.
Degree expected: Feb 2027 (thesis completed). Master's program planned from 2027.

Sep 2005 – Jun 2010 **Specialist Diploma, Law and Economics**, *Novosibirsk State University*, Novosibirsk, Russia

Skills

XR & real-time 3D	Unity, Unreal Engine, AR Foundation, Meta XR SDK (Quest Pro)
Programming	C#, Python, JavaScript (React), CSS
3D content	Blender, Autodesk Maya, Nuke, Fusion 360 (CAD)
UX & research	Figma, Adobe Creative Cloud, user studies (SUS, UEQ)
Tools & methods	Git/GitHub, Scrum

Languages

English	C1
German	B2
Russian	Native